

SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE - 3

Name : HARSH RAJ SHARMA

FOUNDRY - SHOP

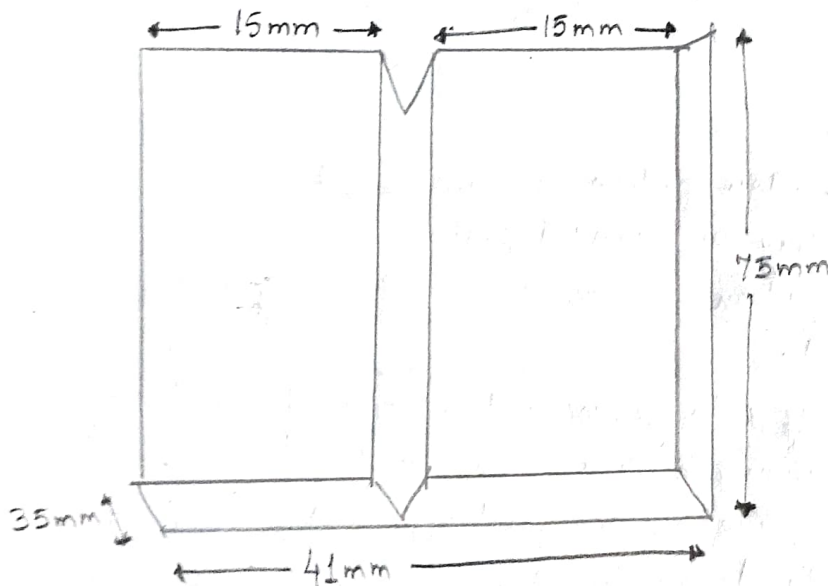
Roll No. : AB-16030

TITLE : To prepare green sand mould of given pattern.

OBJECTIVES :

- Study of material casting process.
- Study of pattern material, types and allowances.
- Moulding process and tools used in moulding.

DRAWING:



Pattern Material :

wood

Scale - 2:1

V-Block wooden pattern

Raw material size :

length - 75mm

width - 45mm

thickness - 35mm

All dia in m.m.

Machine Tool / Hand Tool	Particular	Specification
Type of Moulding sand used	Green Moulding Sand	60-70% fine grain silica sand 20-30% yellow clay
Compositing of Moulding sand	Silica, yellow clay, water, coal dust, & sawdust	5-10% water, 2-5% coal dust, & sawdust
Foundry Tools and Equipment used.	Jowel, Draw, Spikle, Rammer	Hoak & Round Rammer

Sequence of operations :

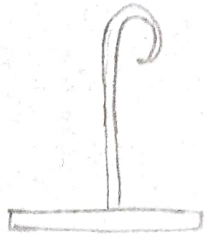
- 1.) Pour the wooden pattern in kerosene oil.
- 2.) Keep the drag in inverted position.
- 3.) Keep the pattern at the centre of the drag and spring some parting sand.
- 4.) Fill the drag with green sand. & ramming is done first with the hook rammer & then with round rammer.
- 5.) Haler are made & some parting sand is sprayed.
- 6.) Cope is placed over the drag runner & riser and made at 1 inch from pattern.

Precautions:

- 1.) Sand should be sieved properly.
- 2.) Pattern should be fixed.
- 3.) Ramming of cope & drag must be done as limits (60-80)% for drag & (55-72)% for cope.
- 4.) The 'v'-pattern must be placed properly into the drag.
- 5.) Parting sand should be sprayed in very small amounts.



Hook Rammer



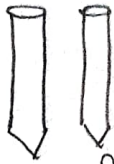
Round Rammer



Strike bar



Vent wire



Runner



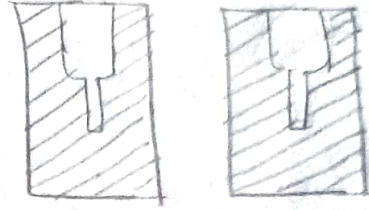
Riser



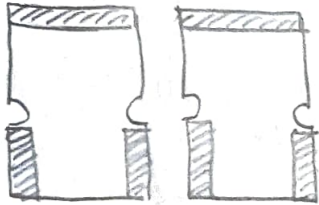
Round Block
Pattern



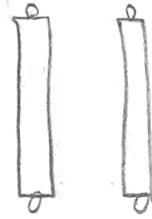
Single T-piece screw
Jack Pattern



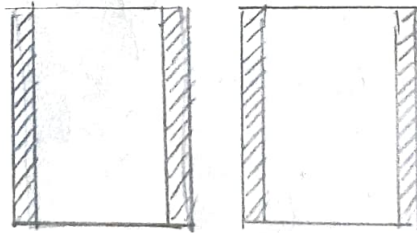
Two piece screw
Stack Pattern



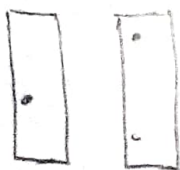
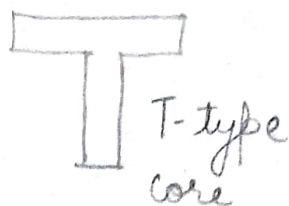
Two piece
T-Type core box



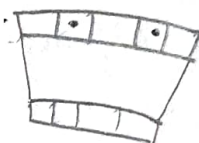
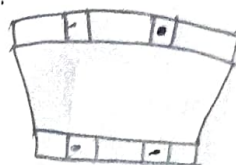
Split piece
pattern



Round Bar Type
Core Box



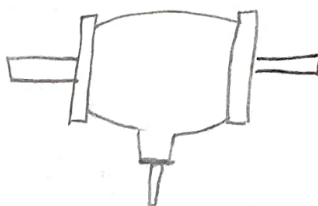
Bar type
Core



also type core



also type core



wheel Volume
Pattern

QUESTIONERIES :

1. Define pattern ?
2. What are the different pattern materials ?
3. List the various type of pattern.
4. Explain various allowances of pattern.
5. What is the difference between Dry Sand Moulding and green sand moulding ?
6. What is Cope and Drag. What is hardness range in cope and Drag in the mould preparing.
7. What do you understand by permeability of mould and how it can be achieved.
8. What is core and core print ?
9. What are different colour code used in pattern ?
10. What is the function of runner and riser ?
11. What is parting line ?
12. Explain the costing defect in short (any five).

① Pattern is the principle tool used in casting process. It may be defined as a model of anything. So constructed that it may be used for forming an impression called mould in damp sand or other suitable material.

② Different types of pattern material are:-

- (i) wood & wood products like laminates.
- (ii) Metal, Brass, Cast iron, Aluminium, white metal
- (iii) Plastics (iv) Plasters (v) Rubbers (vi) Waxes

③ Various patterns are:-

- (i) Single piece pattern (ii) Split pattern (iii) Gated pattern
- (iv) Match plate pattern (v) Cope & drag pattern
- (vi) Shell pattern (vii) Loose piece pattern (viii) Sweep pattern
- (ix) Skeleton pattern (x) Segmental pattern (xi) Built up pattern
- (xii) Boxed up pattern (xiii) Ragged pattern
- (xiv) Left & right hand pattern (xv) Metal pattern

④ The shape & size of a pattern is different from casting, such difference as produced one a pattern called allowance.

(i) Shrinkage allowance:- Pattern create a casting in the mould equal to size and hot matter metal pour into it on cooling to room temperature shrink casting obtain is small.

(ii) Draft Allowance:- When a pattern is drawn from a mould, there is always some possibility of injuring the edges of the metal mould. This danger is greatly decreased if the vertical surface of a pattern are tapered inward. The slight taper on the vertical surface of pattern is known as a draft.

(iii) Machine Allowance:- Rough surface of casting that have to be machined are made to dimensions somewhat over those indicated on the finished working drawings. Extra amount of metal provided on the surface to be machined is called Machine Allowance.

(iv) Distortion or Camber Allowance:- Some casting because of their size, shape & type of metal tend to warp or distort during the cooling period. This is result of uneven shrinkage & is due to uneven metal thickness or to one surface being more exposed than another causing it to cool more rapidly. Shape of the pattern is there bent in the opposite direction to overcome distortion.

(v) Rapping Allowance:- When a pattern is rapped in the mould before it is withdrawn the cavity in the mould is slightly increased. In every case, where castings must be uniform & true to pattern, rapping allowance is provided for by making the pattern slightly smaller than the actual size to compensate for the rapping of the mould.

(5) <u>Green sand moulding</u>	<u>Dry sand moulding</u>
(i) Prepared with natural moulding sands are moisture of silica sand, bonding clay & water. Mould is used in moist.	(i) Different sand is used & mould are dried in an oven before being used for casting.
(ii) Weaker than dry sand.	(ii) Stronger & less susceptible to damage.
(iii) Gives surface defect.	(iii) Doesn't give surface defect.
(iv) Less distortion	(iv) Greater distortion
(v) Lesser dimensional accuracy	(v) Greater dimensional accuracy
(vi) Quick & less expensive	(vi) Takes more time & more expensive
(vii) Less danger of hot tearing of casting	(vii) Castings are susceptible to hot tears.

(6) Cope & Drag refer to the two halves of a pattern or mould used in green sand casting method. The line separating the two halves is called parting line. The upper part is cope & the bottom part is drag.

(7) Permeability is measured by the quantity of air that will pass through a standard specimen of sand under given pressure in a prescribed time. It is property by which it allows the involved gases to escape out. It is achieved by proper sieving of sand & making air holes in it.

⑧ Cases are solids placed in the mould cavity to produce hollow space in casting. Cores are made to rest with their ends in matching depression in the moulds called core points.

⑨ All surfaces of a wooden pattern are coated with shellac to keep out moisture & important parts of a pattern & core box are coloured for identification of their different parts:-

(i.) Surfaces to the left unfurnished are black.

(ii.) Surfaces to be machined are red.

(iii.) Seats for loose pieces are marked by red stripes on a yellow background.

(iv.) Core point are painted yellow.

(v.) Stop-offs are to be marked by diagonal black strips on yellow base.

⑩ The runner & riser are used to leave impression on the mould of drop to where the molten metal is to be poured and to know that the molten metal has filled the cavity. They are to create a path for molten metal in (runner) and out (riser) of the metal mould casting.

⑪ The line separating the upper & lower halves of a mould is called parting line, i.e. the line parting the cope & drag upper & lower halves of the metal is known as the parting line.

(12) Common types of defects found in casting are:-

(i) Shift:- This is an external defect in a casting caused due to care misplacement of top & bottom parts of the casting, usually at a parting line. This can be prevented by ensuring proper alignment of pattern, moulding boxes and correct mounting of patterns.

(ii) Warpage:- It is an unintentional or undesirable deformation in a casting that occurs during or after solidification. Due to different rates of solidification, stresses are set up in adjoining walls resulting in warpage of these areas. Large & flat sections or intersecting sections are particularly prone to warpage.

(iii) Gas Defects:- Gas evolving due to chemical reactions or their decrease in solubility in solidifying the metal causes their entrapment or retaining in the casting & giving rise to pin holes or gas porosity.

(iv) Shrinkage cavity:- It is a void or depression in the casting mainly by uncontrolled & haphazard solidification of the metal. This may also be produced if the pouring temp. is too high. It can be eliminated by directional solidification in mould design & use of chills.

(v) Internal air pockets:- These appear as small holes inside the casting & are caused by pouring boiling metal or by rapid pouring of the molten metal in the mould. Faulty or poor quality metal and excessive moist sand may create air pockets. Correct pouring temp. of the molten metal, right quality of metal & dry sand minimize this defect to a greater extent.